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$$\frac{-1}{2+2i}$$

$$z_1 = -1$$

$$r_1 = \sqrt{(-1)^2 + 0^2} = 1$$

$$\theta_1 = \pi$$

$$z_2 = 2 + 2i$$

$$r_2 = \sqrt{2^2 + 2^2} = \sqrt{8} = 2\sqrt{2}$$

$$\theta_2 = \frac{\pi}{4}$$

$$\Rightarrow r = \frac{r_1}{r_2} = \frac{1}{2\sqrt{2}} = \frac{\sqrt{2}}{4}$$

$$\Rightarrow \theta = \theta_1 - \theta_2 = \pi - \frac{\pi}{4} = \frac{3\pi}{4}$$

$$\Rightarrow z = \frac{\sqrt{2}}{4} \left(\cos\left(\frac{3\pi}{4}\right) + i \sin\left(\frac{3\pi}{4}\right) \right)$$

References